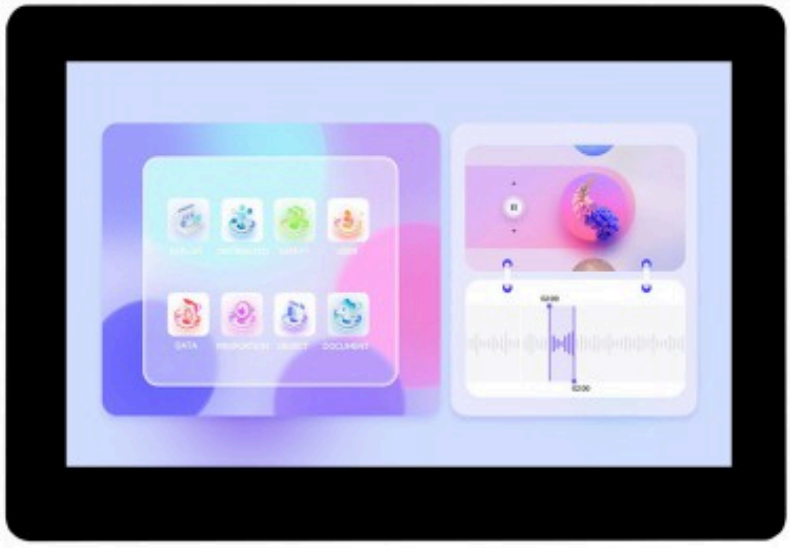
Description Package Content Wiki





WiFi, Bluetooth, Arduino® Compatible, ESP32-S3, LCD (RGB)

ESP32-S3 5inch Display Development Board, 32-Bit LX7 Dual-Core Processor, Up To 240MHz Frequency, Supports WiFi & Bluetooth, With Onboard Antenna, ESP32 With Display

$31.99 - $39.99

ADD TO CART

Version Options

With Touch Function | Without Touch Function

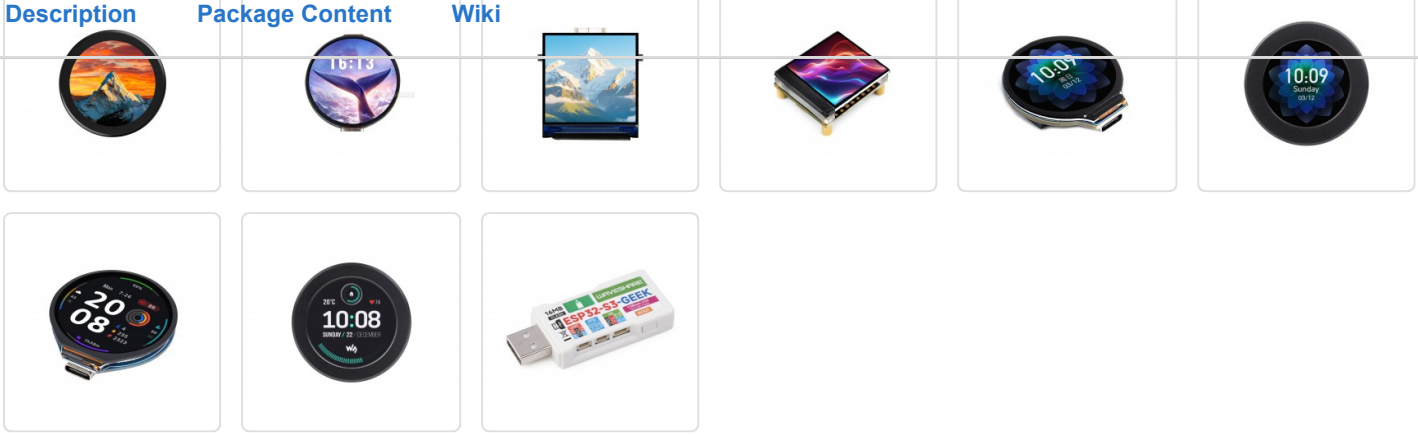
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Select Resolution Package Content Wiki

Standard Ver (800×480 pixels) | Upgraded Ver (1024×600 pixels)

Related Products:



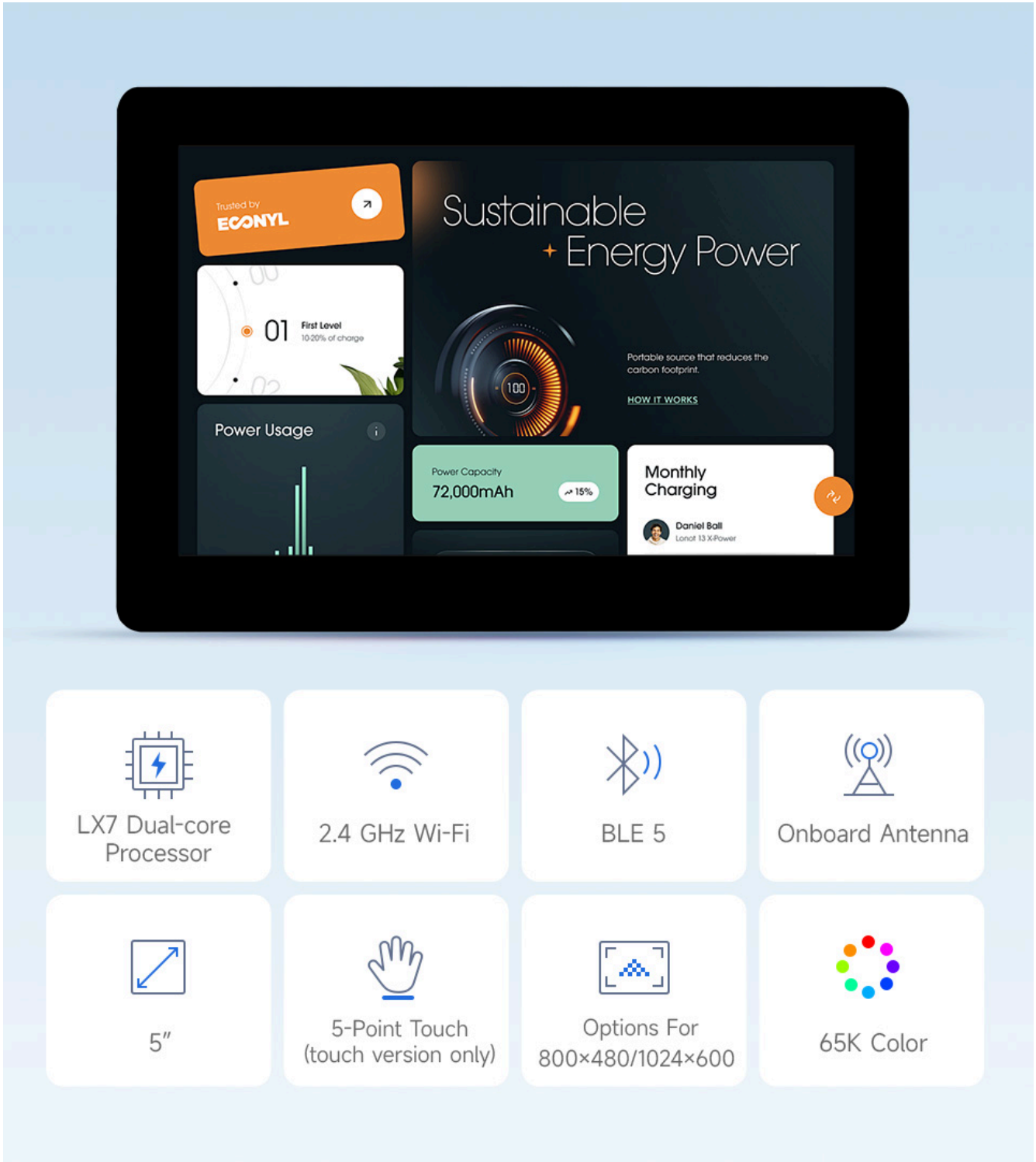


ESP32-S3 5inch Display Development Board, 32-Bit LX7 Dual-Core Processor, Up To 240MHz Frequency, Supports WiFi & Bluetooth, With Onboard Antenna, Options For 800×480/1024x600 Display Resolution And Touch Function

ESP32-S3 5″ LCD Development Board

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Integrates RGB Interface LCD And 5-Point Capacitive Touch (Optional)



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Version Options

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ESP32-S3-LCD-5



Standard version, without touch

800×480 pixels, 160° viewing angle

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ESP32-S3-LCD-5B



Upgraded version, without touch

1024×600 pixels, 178° viewing angle

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ESP32-S3-Touch-LCD-5



Standard version, with touch function

800×480 pixels, 160° viewing angle

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ESP32-S3-Touch-LCD-5B



Upgraded version, with touch function

1024×600 pixels, 178° viewing angle

Features

This product is a microcontroller development board with 2.4GHz WiFi and BLE 5 support, integrates high-capacity Flash and PSRAM. Onboard 5inch LCD screen can smoothly

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run GUI programs such as LVGL. Combined with various peripheral interfaces, suitable for the quick development of the HMI and other ESP32-S3 applications

- Equipped with Xtensa 32-bit LX7 dual-core processor, up to 240MHz main frequency
- Supports 2.4GHz Wi-Fi (802.11 b/g/n) and Bluetooth 5 (LE), with onboard antenna
- Built in 512KB SRAM and 384KB ROM, with onboard 16MB Flash and 8MB PSRAM
- Onboard 5inch LCD display, options for 800×480/1024x600 resolution, 65K color
- Supports capacitive touch control via I2C interface, 5-point touch with interrupt support (for touch version only)
- Onboard voltage regulator, supports 7~36V wide range power supply
- Onboard RTC chip and battery holder, ensuring time data is not lost when power loss (RTC battery required)
- Onboard CAN, RS485, I2C interface, TF card slot, MX1.25 2P battery header, and digital isolated IO interfaces, etc.
- Supports passive and active digital input, with bi-directional optocoupler isolation

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- Supports digital output with optocoupler isolation, provides higher driving capability with up to 450mA sinking current
- Onboard LED indicators for indicating the power and battery recharging status
- Supports flexible clock, module power supply independent setting and other controls to realize low power consumption in different scenarios

Application Scenarios

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Human-machine Interface

The Human-machine Interface (also known as the user interface) is the medium of interaction and information exchange between the system and the user, it realizes the transformation between the internal form of information and the form acceptable to human beings.



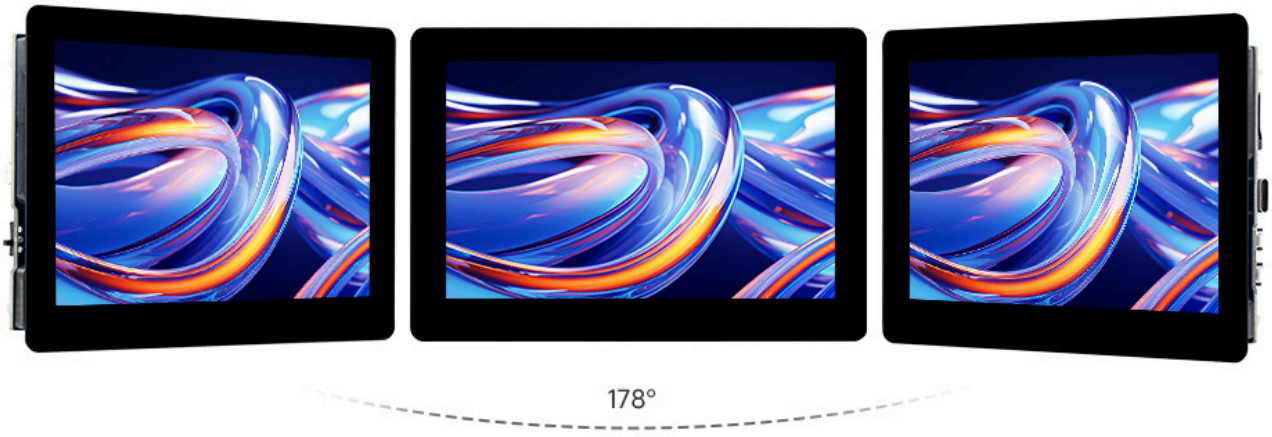
LVGL GUI Development

LVGL is a free, open-source graphics library that provides everything you need to create embedded GUI with the easy-to-use graphical elements, beautiful visual effects and low memory requirement.

IPS Display Panel

Excellent Display Performance, 178° Wide Viewing Angle *

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* 178° viewing angle for the Upgraded Version, and 160° for the Standard Version

Capacitive 5-Point Touch (Optional)

Optional For Touch Function Via I2C Interface, Supports 5-Point Touch For Flexible Operation And Smooth Multi-Point UI Interaction

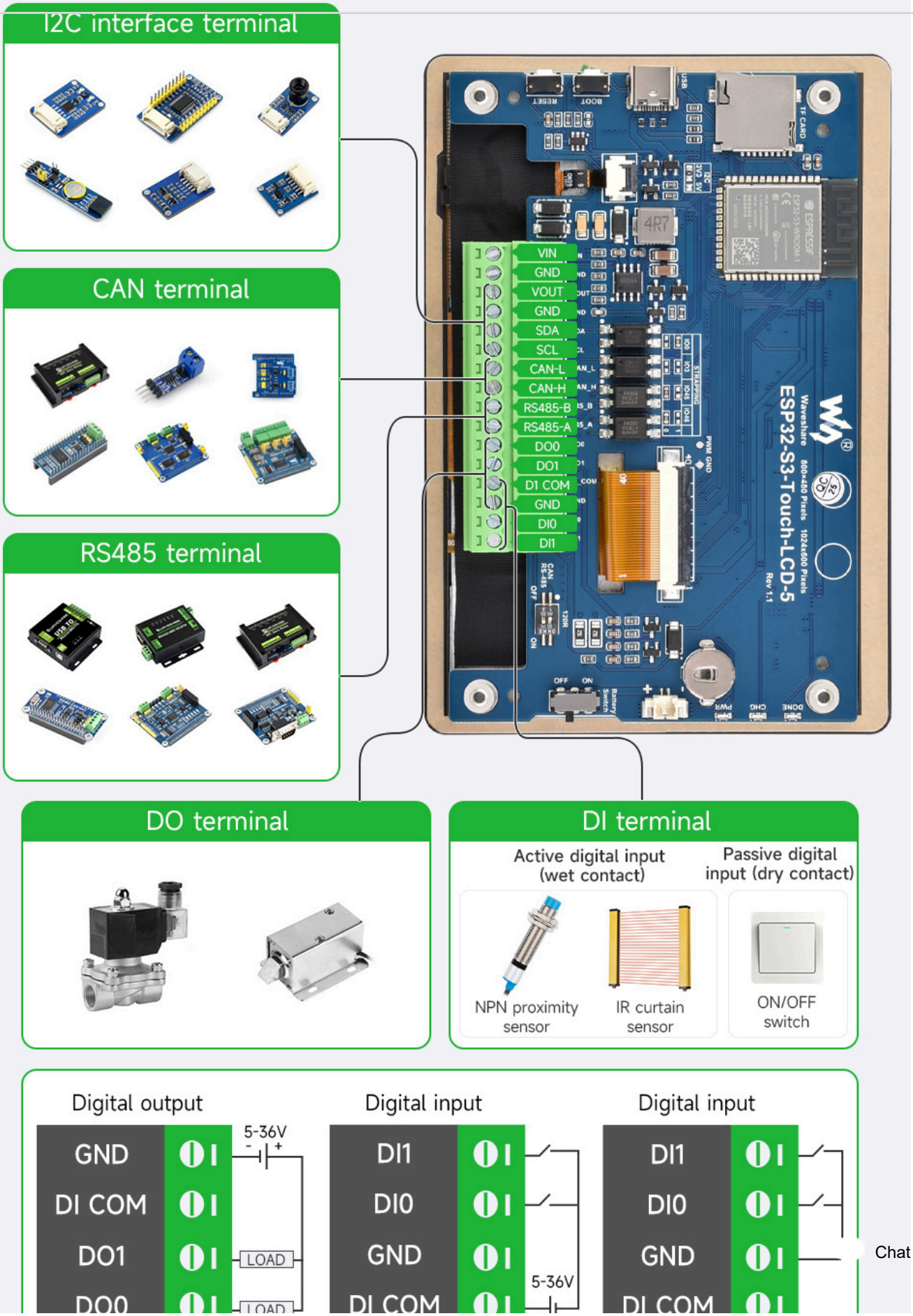
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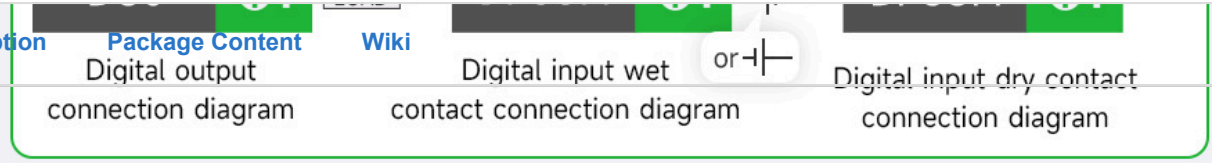
Supports Multiple Peripherals

Supports The Expansion Of Multiple Peripherals Via CAN, RS485, I2C And Isolated IO Interfaces

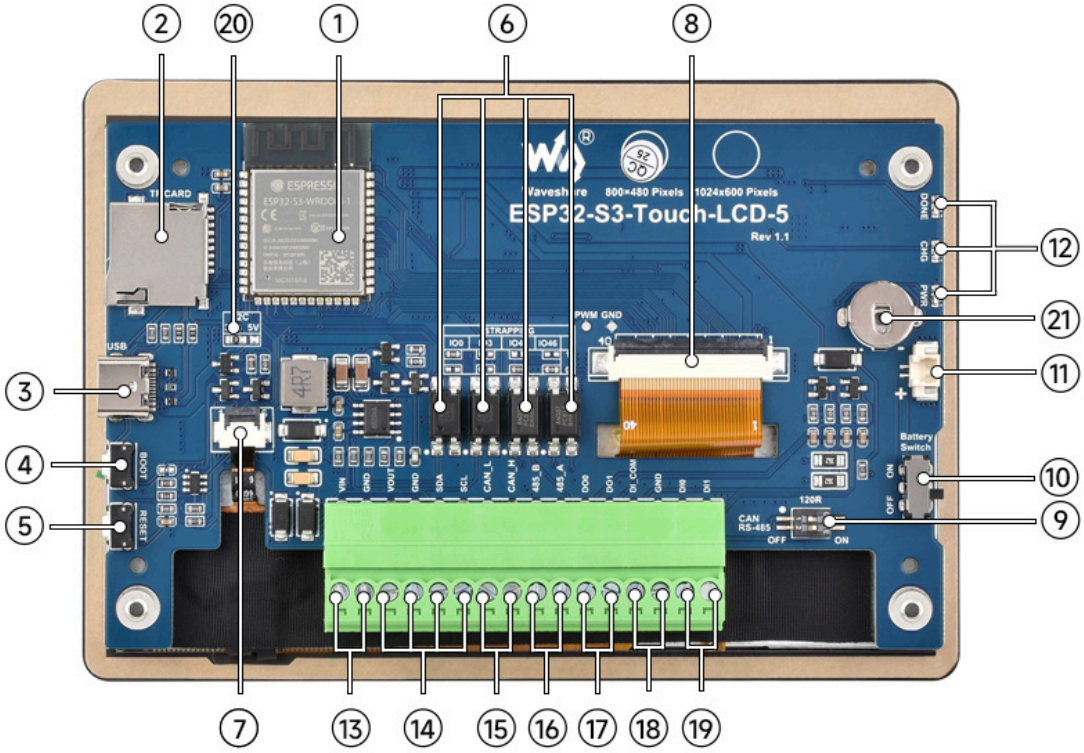
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I2C interface terminal
CAN terminal
RS485 terminal
VIN
GND
VOUT
GND
SDA
SCL
CAN-L
CAN-H
RS485-B
RS485-A
DO0
DO1
DI COM
GND
DI0
DI1
ESP32-S3-Touch-LCD-5
DO terminal
DI terminal
Active digital input (wet contact)
Passive digital input (dry contact)
NPN proximity sensor
IR curtain sensor
ON/OFF switch
Digital output
Digital input
Digital input
5-36V
GND
DI COM
DO1
DO0
LOAD
LOAD
DI1
DI0
GND
DI COM
5-36V
DI1
DI0
GND
DI COM

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What's On Board



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1. **ESP32-S3-WROOM-1-N16R8**

   The SoC with WiFi and Bluetooth, up to 240MHz operating frequency, integrated 8MB PSRAM and 16MB Flash
2. **TF card slot**
3. **USB Type-C port**
4. **BOOT button**

   Press and hold while powering on for program burning
5. **RESET button**
6. **Optocoupler isolation**
7. **Touch panel connector**
8. **5inch display panel connector**
9. **RS485 and CAN terminal resistors selection**

   close by default
10. **Battery power supply ON/OFF**
11. **3.7V single lithium battery GH1.25 header**
12. **Status Indicators**

    DONE: Lithium battery charging completed indicator

    CHG: Lithium battery charging indicator

    PWR: Power supply indicator

    (the CHG will blink and DONE will light on when connecting with DC power supply and without lithium battery connected or battery switch is OFF)

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13. **7-36V DC power supply terminal**
14. **I2C interface terminal**

    VOUT pin output 5V/3.3V (switching by onboard resistor)
15. **CAN interface terminal**
16. **RS485 interface terminal**
17. **Digital output**

    5~36V, open-drain output, output load: 450mA per channel (MAX)
18. **Input signal common terminal**

    NC: dry contact passive input

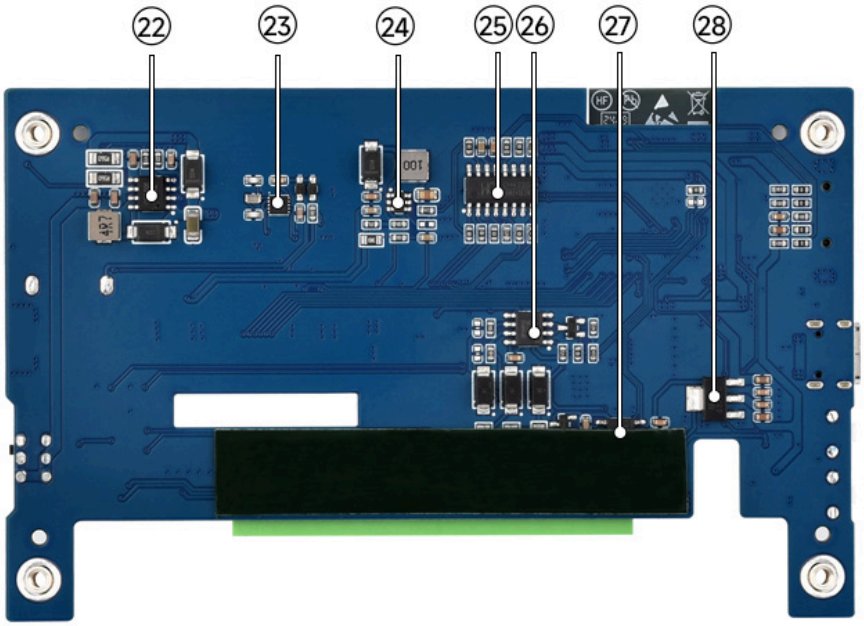
    Connect to power supply positive: digital input low active, for NPN wet contact active input, 5V-36V DC

    Connect to power supply negative: digital input high active, for PNP dry contact active input, 5V-36V DC
19. **Digital input**

    5~36V, supports passive / active input (NPN or PNP)
20. **I2C VOUT output selection**

    for switching output valtage and I/O level of I2C VOUT, 3.3V by default
21. **RTC battery holder**

    Compatible with CR927 battery



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22. **CS8501**
    Lithium battery recharge manager
23. **PCF85063**
    RTC chip for providing accurate time control
24. **AP3032KTR-G1**
    Display backlight boost converter
25. **CH422G**
    IO expansion chip

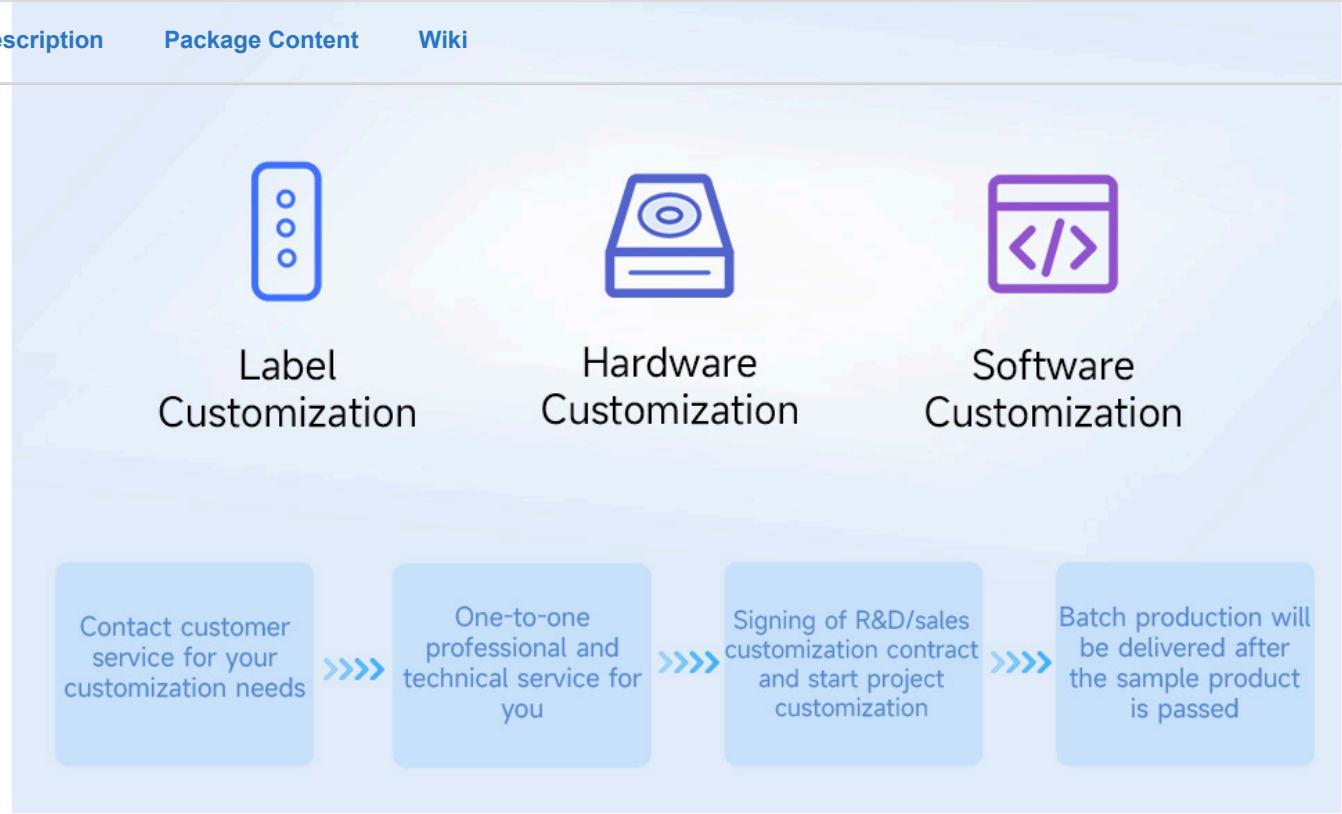
26. **SP3485**
    RS485 transceiver
27. **TJA1051T/3/1J**
    CAN transceiver
28. **SGM2212-3.3**
    800mA low-noise LDO

Support Batch Customization

Support Software And Hardware Customization

Including Hardware Interface, LOGO, Label, Case And Web Page, Etc.

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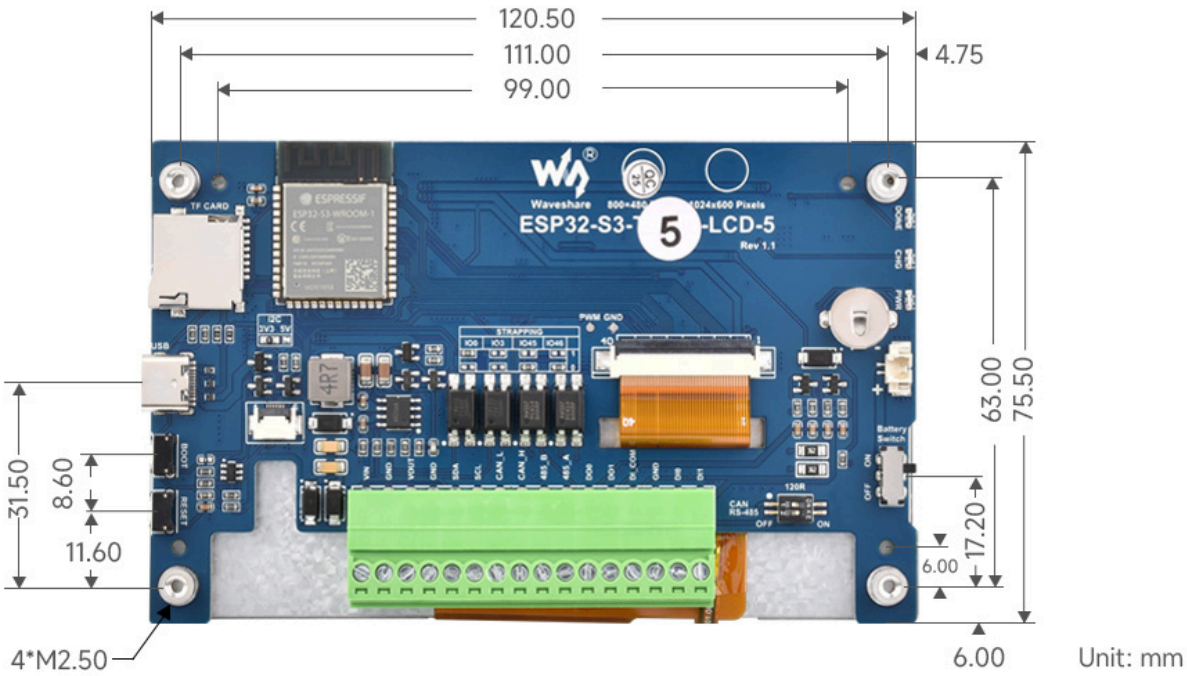


Outline Dimensions

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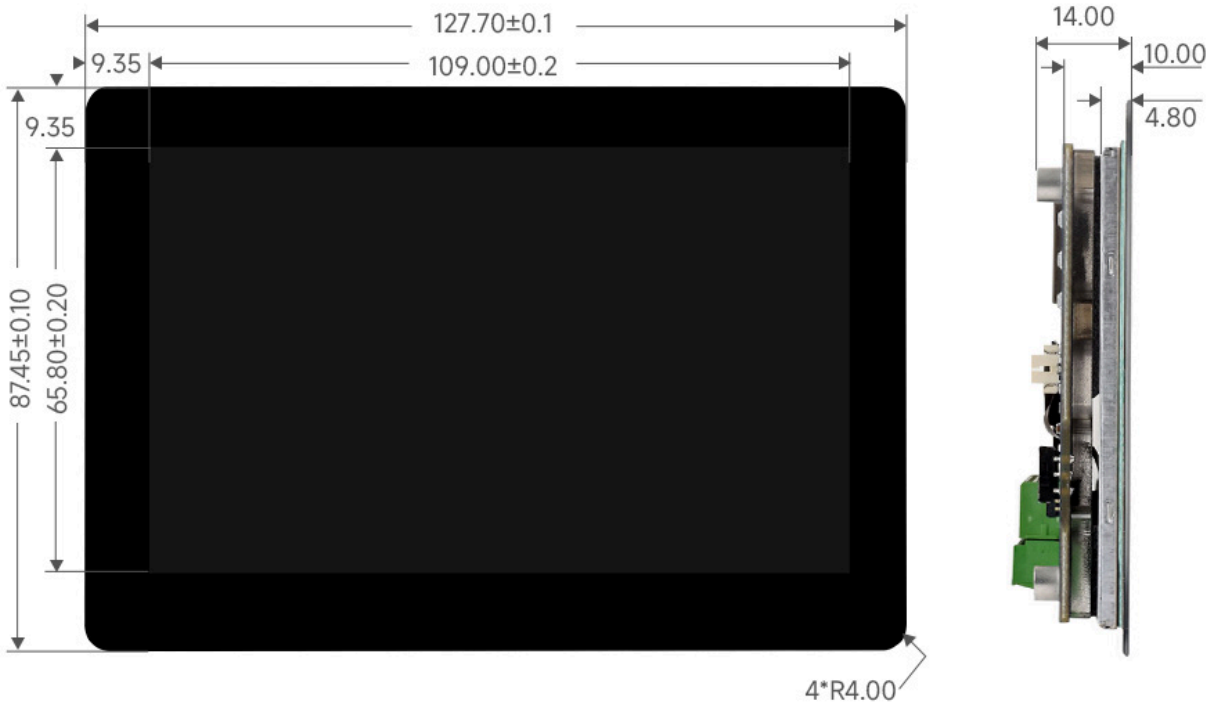
Without Touch Version

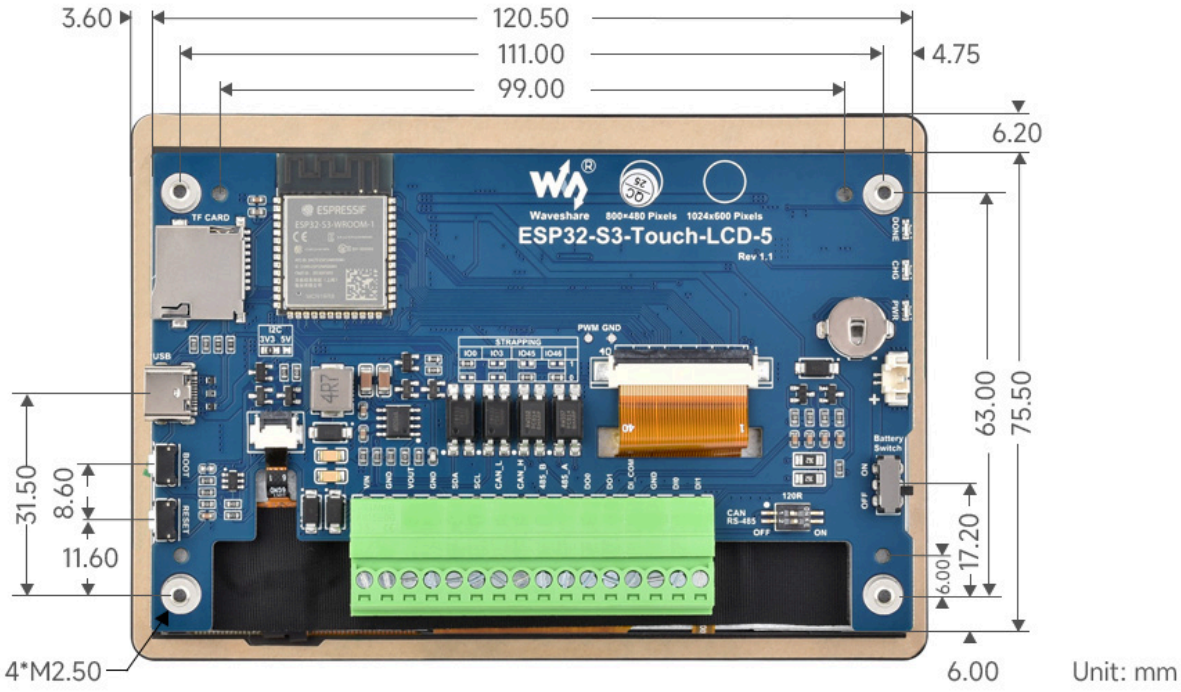




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With Touch Version





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Resources & Services

WAVESHARE ORIGINAL

PROFESSIONAL TECH SUPPORT

RICH RESOURCES MANUAL & DEMO

Wiki: www.waveshare.com/wiki/ESP32-S3-Touch-LCD-5

ESP32 LCD Selection

MODEL	BASIC PARAMETERS			LCD PARAMETERS			COMMUNICATION INT		
	MCU	FLASH	PSRAM	RESOLUTION	VIEWING ANGLE	TOUCH TYPE	CAN	RS485	I2C
 ESP32-S3-Touch-LCD-4 Without Touch / With Touch	ESP32-S3	16MB	8MB	480 × 480	160°	capacitive	√	√	√
 ESP32-S3-Touch-LCD-4.3B Standard Ver. / Case Ver.	ESP32-S3	16MB	8MB	800 × 480	160°	capacitive	√	√	√

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Description	Package Content	Wiki							
 ESP32-S3 4.3″ LCD Without Touch / With Touch	ESP32-S3	16MB	8MB	800 × 480	160°	capacitive	√	√	√
 ESP32-S3 5″ LCD Without Touch / With Touch	ESP32-S3	16MB	8MB	800 × 480	160°	capacitive	√	√	√
 ESP32-S3 5″ LCD (B) Without Touch / With Touch	ESP32-S3	16MB	8MB	1024 × 600	178°	capacitive	√	√	√
 ESP32-S3-Touch-LCD-7 Without Touch / With Touch	ESP32-S3	16MB	8MB	800 × 480	170°	capacitive	√	√	√

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Description		Package	Content	Wiki						
ESP32-S3-Touch-LCD-7B Without Touch / With Touch		ESP32-S3	16MB	8MB	1024 × 600	170°	capacitive	√	√	√

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